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-force of attraction b/w two electric charges and two parallel conductors

i) force b/w two electric charge $f = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$

-force b/w two parallel conductors $f = \frac{\mu_0 I_1 I_2 l}{2\pi d}$

$q_1 = 3 \times 10^{-6}$

$q_2 = 5 \times 10^{-6}$

$r = 3 \text{ cm}$

$I_1 = 5$

$I_2 = 3$

$l = 5 \text{ cm}$

$d = 3 \text{ cm}$

(i) force b/w two electric charges .

$f = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$

$= \frac{(3 \times 10^{-6})(5 \times 10^{-6})}{4 \times 3.14 \times 8.85 \times 10^{-12} \times (3 \times 10^{-2})^2}$

$= \frac{1.5 \times 10^{-11}}{1.000856 \times 10^{-13}}$

$= 1.4987 \text{ N}$

(ii) force b/w two parallel conductors

$f = \frac{\mu_0 I_1 I_2 l}{2\pi d}$

$= \frac{(4 \times 3.14 \times 10^{-7})(5)(3)(5 \times 10^{-2})}{2 \times 3.14 \times 3 \times 10^{-2}}$

$= \frac{9.42 \times 10^{-7}}{0.01884} = 5 \times 10^{-5} \text{ N}$

$= 0.0001$